CSS142P: Modeling and Simulation Theory

Project Instructions

Simulation Case Study and Investigation

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1 Project description

The project is a case study and investigation using modeling and simulation. Each team shall select a real or realistically motivated system, define a focused research or decision problem, develop a simulation model, conduct experiments, analyze the results, and state a recommendation supported by evidence.

The project must use a dynamic or stochastic simulation model. A dashboard, visualization, deterministic computation, or machine-learning accuracy comparison by itself is not sufficient. Acceptable simulation paradigms include discrete-event simulation, agent-based simulation, Monte Carlo simulation, system dynamics, and justified hybrid approaches.

Teams should normally consist of three to four students unless the instructor announces a different arrangement.

2 Project objectives

At the end of the project, students should be able to:

1. define a research or decision question suitable for simulation;
2. establish the system boundary, assumptions, inputs, state variables, events, rules, and outputs;
3. select and justify an appropriate modeling and simulation approach;
4. implement and verify a computational model;
5. validate the model for its intended purpose;
6. design experiments and quantify stochastic uncertainty;
7. perform sensitivity or robustness analysis;
8. interpret results without making claims beyond the evidence; and
9. prepare a reproducible technical report and presentation.

3 Conference tracks

Every team must identify one primary track for its project. An optional secondary track may be named when the work is genuinely cross-track. The following names are taken directly from the official 2024 Winter Simulation Conference program and proceedings. They are not locally created project categories.

1. Agent-Based Simulation
2. Analysis Methodology
3. Commercial Case Studies
4. Complex and Resilient Systems
5. Data Science for Simulation
6. Simulation as Digital Twin
7. Simulation Optimization

Teams shall use the official proceedings index to review the papers and sessions within the selected track:

`https://informs-sim.org/wsc24papers/by_area.html`

The selected track does not replace the research question. It identifies the conference research community to which the project is most closely related.

4 Minimum technical requirements

Every project must contain the following:

1. one clearly stated research or decision question;
2. one baseline scenario and at least two alternative scenarios or policies;
3. at least three uncertain inputs or factors;
4. a conceptual model prepared before implementation;
5. a working computational simulation model;
6. at least four verification checks;
7. at least two validation approaches;
8. a justified run length, warm-up procedure when applicable, and number of replications;
9. point estimates and uncertainty intervals for the principal stochastic outputs;
10. at least one sensitivity or robustness analysis;

11. at least six credible references, including at least two peer-reviewed papers and at least one paper from the supplied reading set; and
12. a reproducibility package containing the model, data or data generator, configuration, dependencies, random-seed procedure, instructions, and output needed to reproduce the principal result.

5 Required chapters of the project proposal

The proposal shall be organized using the following numbered chapters. The proposal should normally be four to six pages excluding references and appendices.

Chapter 1. Project title and team information

Provide the project title, names of all members, student numbers, section, assigned team roles, and proposed primary WSC track. Include an optional secondary WSC track only when necessary.

Chapter 2. Case background and problem context

Describe the system to be studied, its stakeholders, its current process or behavior, and the practical importance of the problem. Identify the decision maker or intended user of the results. State why simulation is appropriate for the case.

Chapter 3. Research question, objectives, and scope

State one primary research or decision question. Enumerate the specific objectives. Define the system boundary, time horizon, unit of analysis, performance measures, inclusions, and exclusions. Identify the baseline and the proposed alternatives.

Chapter 4. Conference track and review of related literature

Name the selected official WSC track and explain the alignment of the proposed work with papers or sessions in that track. Summarize the most relevant literature. Identify the modeling methods, findings, or research gaps that inform the proposed investigation. Cite at least four credible sources in the proposal.

Chapter 5. Proposed conceptual model

Present the initial system-boundary diagram and an appropriate process-flow, state-transition, influence, or agent-interaction diagram. Enumerate the entities or agents, resources, state variables, events, queues, rules, parameters, inputs, outputs, and assumptions that apply to the chosen paradigm.

Chapter 6. Data and input-modeling plan

Identify the required data, expected data sources, units, collection or generation procedure, authorization, preprocessing, and quality limitations. Distinguish observed, estimated, assumed, and calibrated values. Explain how input distributions or parameter values will be selected. If synthetic data will be used, describe the generator and the limits this places on the conclusions.

Chapter 7. Proposed simulation methodology and implementation plan

State the simulation paradigm, software or programming environment, model time unit, initialization procedure, stopping condition, and random-seed policy. Explain why the selected approach is suitable. Provide a preliminary model architecture or pseudocode and identify required libraries or tools.

Chapter 8. Verification and validation plan

Enumerate the verification procedures that will be used, such as deterministic cases, event traces, conservation checks, boundary tests, assertions, unit tests, manual walkthroughs, or independent code review. Enumerate the proposed validation approaches, such as face validation, comparison with held-out observations, historical reproduction, output-distribution comparison, extreme-condition testing, pattern validation, or cross-model comparison.

Chapter 9. Experimental and analysis plan

Define the baseline, alternatives, experimental factors, factor levels, response measures, run length, warm-up treatment, proposed replication method, and comparison procedure. State how uncertainty intervals will be calculated. Identify the planned sensitivity or robustness analysis and the criteria for recommending one alternative.

Chapter 10. Work plan, responsibilities, and risks

Provide the project schedule and milestones. Assign responsibilities for literature review, data preparation, model development, verification, validation, experimentation, analysis, writing, and presentation. Identify technical, data, schedule, and ethical risks together with mitigation actions.

Chapter 11. References

List all sources cited in the proposal using one consistent citation style. Include stable links or digital object identifiers when available.

Proposal appendices, when needed

Appendices may contain the initial evidence table, data dictionary, interview guide, extended diagrams, preliminary code, or other supporting material. Appendices do not replace explanations required in Chapters 1–11.

6 Required chapters of the final project

The final report shall be organized using the following numbered chapters. The report should normally be eight to twelve pages excluding references and appendices unless the instructor provides a different format.

Chapter 1. Introduction and case background

Introduce the system, stakeholders, practical problem, motivation, and role of simulation. State the primary research or decision question and summarize the contribution of the project.

Chapter 2. Objectives, scope, and conference-track alignment

Enumerate the completed objectives. Define the final system boundary, time horizon, performance measures, baseline, alternatives, inclusions, and exclusions. Name the official WSC track and explain the final alignment of the study with that track.

Chapter 3. Review of related literature

Synthesize the literature used to design and interpret the study. Compare relevant modeling approaches and findings. Explain how the project adopts, adapts, tests, or extends ideas from prior work. At least one supplied WSC paper must be discussed substantively.

Chapter 4. Conceptual model and assumptions

Present the final system-boundary and model diagrams. Define all important entities or agents, resources, state variables, events, queues, behaviors, interactions, equations, parameters, inputs, and outputs. Enumerate the assumptions, their evidence or rationale, and their likely effect on the results.

Chapter 5. Data and input modeling

Describe the data sources, authorization, units, sampling or generation procedure, preprocessing, missing-data treatment, fitted distributions, parameter estimation, and data limitations. Include a parameter-provenance table that marks each value as observed, estimated, assumed, or calibrated. Distinguish calibration data from validation data when possible.

Chapter 6. Model implementation

Describe the software, libraries, model architecture, time advance, initialization, stopping rules, event or agent logic, seed policy, and important implementation decisions. Provide enough detail for a knowledgeable reader to understand how the conceptual model became an executable model.

Chapter 7. Model verification

Report at least four verification checks and their results. Show how expected behavior was determined for each check. Include defects found, their causes, and the corrections made. Merely stating that the model ran successfully is not verification.

Chapter 8. Model validation

Report at least two complementary validation approaches and their results. Explain the validation measures and acceptance criteria. Conclude whether the model is adequate, conditionally adequate, or inadequate for the stated purpose. Do not claim that the model is universally valid.

Chapter 9. Experimental design

Define the baseline, alternatives, factors, factor levels, response measures, run length, warm-up treatment, replication count, seed allocation, variance-reduction method when applicable, and statistical comparison procedure. Justify the replication count using precision, pilot results, or a sequential stopping rule.

Chapter 10. Results

Present the principal results using appropriately labeled tables and figures. Report point estimates, uncertainty intervals, practical effect sizes, and comparisons among scenarios. Explain relevant model behavior and mechanisms rather than reporting output values alone.

Chapter 11. Sensitivity and robustness analysis

Challenge the principal conclusion by varying important input values, distributional assumptions, structural assumptions, or scenario definitions. Identify which factors materially affect the result and the conditions under which the preferred alternative changes.

Chapter 12. Discussion, limitations, and responsible practice

Interpret the results in relation to the case and literature. Discuss trade-offs, generalizability, data limitations, model limitations, possible bias, privacy, ethics, security, environmental impact when relevant, and the consequences of a wrong recommendation. Disclose permitted generative-AI assistance according to current course and university policy.

Chapter 13. Conclusions and recommendations

Answer the research question directly. State the recommended action, the evidence supporting it, the conditions under which it applies, and the principal uncertainty. Identify the next data or modeling improvement that would most strengthen the decision.

Chapter 14. Reproducibility and team contributions

Describe the contents of the reproducibility package and the exact procedure for reproducing the principal result. State software versions and computational requirements. Enumerate the verified contributions of each team member.

Chapter 15. References

List every cited source using one consistent citation style. Provide stable links or digital object identifiers when available.

Final project appendices

Include supporting material that is necessary for audit but too detailed for the main report, such as extended parameter tables, test logs, validation calculations, supplemental results, code excerpts, and data dictionaries. The complete executable files shall be submitted separately in the reproducibility package.

7 Other final deliverables

In addition to the final report, submit:

1. the complete executable simulation model;
2. the data, a permitted subset, or a reproducible data generator;
3. a README with installation and execution instructions;
4. an environment or dependency specification;
5. configuration files and the documented random seeds;
6. the output used to create the principal result;
7. an eight- to ten-minute presentation or a poster, as announced by the instructor; and
8. an individual contribution and reflection statement from each member.

8 Assessment rubric

Criterion	Points
Case framing, objectives, scope, and literature	12
Conceptual model, assumptions, data, and input modeling	15
Implementation and verification	15
Validation and model credibility	15
Experimental design and uncertainty treatment	15
Results, sensitivity analysis, and interpretation	15
Reproducibility, responsible practice, and contributions	8
Report and presentation quality	5
Total	100

9 Supplied conference-paper reference set

The `Papers` folder contains the following twelve open-access papers from the official 2024 Winter Simulation Conference proceedings. Track and session names below reproduce the conference program terminology.

1. Enlu Zhou, "Data-driven Simulation Optimization in the Age of Digital Twins: Challenges and Developments." Track: Advanced Tutorials. `https://informs-sim.org/wsc24papers/inv153.pdf`

2. Young Jae Jang et al., "Concept of Digital Twins for Autonomous Manufacturing through Virtual Learning and Commissioning." Track: Advanced Tutorials. `https://informs-sim.org/wsc24papers/inv191.pdf`

3. Hao He, Xueying Liu, Chris J. Kuhlman, and Xinwei Deng, "A Framework of Digital Twins for Modeling Human-Subject Word Formation Experiments." Track: Agent-Based Simulation. Session: Agent-Based Modeling and Emerging Trends. `https://informs-sim.org/wsc24papers/con170.pdf`

4. Yuhang Wu, Yingfei Wang, Chu Wang, and Zeyu Zheng, "LLM Enhanced Machine Learning Estimators for Classification." Track: Analysis Methodology. Session: Machine Learning and AI. `https://informs-sim.org/wsc24papers/inv174.pdf`

5. Linyun He, Luke Rhodes-Leader, and Eunhye Song, "Digital Twin Validation with Multi-epoch, Multi-variate Output Data." Track: Analysis Methodology. Session: Digital Twins and Calibration. `https://informs-sim.org/wsc24papers/inv203.pdf`

6. Peter Haas, "Tutorial: Artificial Neural Networks for Discrete-event Simulation." Track: Advanced Tutorials. `https://informs-sim.org/wsc24papers/inv218.pdf`

7. Hiromitsu Hattori, Arata Kato, and Mamoru Yoshizoe, "Integrating Large Language Models into Agent Models for Multi-Agent Simulations: Preliminary Report." Track: Agent-Based Simulation. Session: Agent-Based Modeling and Emerging Trends. `https://informs-sim.org/wsc24papers/con310.pdf`

8. Haoting Zhang, Jinghai He, Jingxu Xu, Jingshen Wang, and Zeyu Zheng, “Enhancing Language Model with Both Human and Artificial Intelligence Feedback Data.” Track: Analysis Methodology. Session: Machine Learning and AI. https://informs-sim.org/wsc24papers/con317.pdf
9. Ruhollah Jamali and Sanja Lazarova-Molnar, “A Comprehensive Framework for Data-Driven Agent-Based Modeling.” Track: Data Science for Simulation. Session: Data-Driven Modeling and Simulation Frameworks. https://informs-sim.org/wsc24papers/inv188.pdf
10. Abdolreza Abhari, “Towards New Simulation Models For DL-based Video Streaming In Edge Networks.” Track: Data Science for Simulation. Session: Machine Learning and Data Utilization in Simulation. https://informs-sim.org/wsc24papers/con334.pdf
11. Lourdes Murphy, Nelson Alfaro Rivas, and Yusuke Legard, “Streamlining Security: Using Agent-Based Simulation to Reduce Congestion and Enhance Campus Security Screening.” Track: Commercial Case Studies. Session: Simulation to Improve Customer Experience. https://informs-sim.org/wsc24papers/cea131.pdf
12. Ranjan Pal, Rohan Sequeira, and Sander Zeijlemaker, “How Hard is it to Estimate Systemic Enterprise Cyber-Risk?” Track: Complex and Resilient Systems. Session: System Resilience and Cyber Risk. https://informs-sim.org/wsc24papers/inv125.pdf

Official proceedings index:

https://informs-sim.org/wsc24papers/by_area.html

10 Conference submission information

External submission is optional, is not required for the course grade, and must be approved by the instructor and all team members.

The 2026 IEEE/ACM International Symposium on Distributed Simulation and Real Time Applications lists poster, demonstration, and industrial case-study submissions. Its official poster page listed July 25, 2026 as the submission deadline when checked on June 23, 2026.

- DS-RT 2026 poster and case-study call
- DS-RT 2026 submission guidelines
- DS-RT 2026 call for papers
- WSC 2026 call for papers
- WSC 2026 author and submission information

Conference deadlines and policies may change. The official conference website is authoritative.

11 Academic integrity and responsible use

All borrowed prose, code, data, models, diagrams, and figures must be cited and used according to their licenses. Do not submit confidential data, personal identifiers, passwords, or API keys. Use only data that the team is authorized to access. Any permitted use of generative AI must be disclosed and verified according to current course and university policy. Every team member remains responsible for the submitted work.